Loka Surya Prakash Pentakota

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PROFESSIONAL SUMMARY

Results-driven Software Engineer specializing in Distributed Systems. Demonstrated expertise in engineering high-reliability systems and production services utilizing Javascript, Python, React.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, React Native, Expo, TailwindCSS, React

Backend: Node.js, Express.js, Flask, FastAPI, REST APIs, WebSockets

AI/ML: Retrieval-Augmented Generation (RAG), Agentic AI, LLM Orchestration, VLMs, Computer Vision, OCR, YOLO, SAHI, Artificial Intelligence

Systems: Distributed Systems, MAVLink, WebRTC, GStreamer, ZeroMQ

Databases/Cloud: MongoDB, PostgreSQL, Redis, AWS, GCP

WORK EXPERIENCE

JulleyOnline Private Limited India

Platform Engineer *April 2026 – September 2026*

PMS Integration Hub

- Architected a centralized integration hub using adapter and registry patterns, which cut development time for new third-party integrations by 42% and standardized data synchronization.
- Integrated Gmail, Zoho, Microsoft Outlook, Naukri, and ESSL through reusable connector workflows for seamless authentication and data exchange, improving system throughput by 28% under peak production load.

Drone GCS & Fleet Management Platform

- Engineered a distributed ArduPilot Ground Control Station (GCS) using React, Node.js, Python, FastAPI, pymavlink, MAVLink, and ZeroMQ for real-time drone monitoring and control, boosting query execution efficiency by 35% across 500K+ records.
- Orchestrated the end-to-end GCS command and control stack, enabling real-time mission planning and robust reconnect handling that maintained a 99.9% operational uptime across drone fleets.
- Designed a ZeroMQ + WebSocket telemetry pipeline for low-latency vehicle-state updates across multiple clients, achieving 99.9% uptime SLA with zero-downtime CI/CD deployments.

AI-Powered Engineering Drawing Tag Extraction

- Spearheaded an Artificial Intelligence pipeline in Python combining YOLO, SAHI, OCR, OpenCV, and Gemini to extract equipment tags from complex P&ID drawings, increasing tag extraction accuracy by 38% while reducing manual review time.
- Implemented a Retrieval-Augmented Generation (RAG) based symbol knowledge system by extracting symbol-label mappings from engineering reference PDFs and creating vector representations for retrieval.
- Combined retrieved engineering context with visual detection, dimensions, scale, and orientation to associate detected components with their corresponding symbols and tags.

Vertx Remote

Founding Full-Stack Engineer *August 2024 – February 2026*

Vertx Flow | *AI Fundraising Platform*

- Engineered 5+ distinct AI workflows for investor matching and founder evaluation, improving match quality by 42% while reducing manual review time by 10 hours per week.
- Engineered fundraising pipelines, outbound outreach, shareable pitch links, and engagement tracking mechanisms.

Vertx Startin | *Founder-Investor Network*

- Spearheaded and launched a full-stack founder-investor platform using React.js, Node.js, Express.js, and MongoDB on AWS, scaling to support 1,000+ daily active users within 6 months of launch.
- Launched a semantic search and matching engine that improved founder-investor discovery accuracy by 42%, driving a 42% uplift in user engagement on the platform.

EDUCATION

Vignan's Institute Of Information Technology

Vignan's Institute Of Information Technology *Visakhapatnam, India 2021 – 2025 Bachelor of Technology in Computer Science an*